

increasing checkpoint payload variability

Fixed-size State

Variable-size State

Iterative

Task-parallel

Pipeline

Irregular

CoMD 2K
LULESH 5K
miniFE 3K
arrays, structs
loop boundary ckpt

Chameleon 5K
DPLASMA 5K
tile matrices
task boundary ckpt
ABFT possible

Palabos 50K
Nektar++ 3K
SU2 400K
staged phases
inter-stage ckpt

AMG 30K
miniVite 3K
graph / multigrid
fixed partitions

LAMMPS 300K
WarpX 150K
Nyx 15K
particles, AMR
dynamic payload

CLAMR 15K
deal.II 2K
adaptive mesh
topology changes

RAxML-NG 5K
ExaML 2K
search iteration
variable tree size

SAMRAI 200K
Uintah 3K
AMReX 150K
dynamic refinement
level-dependent ckpt

POSIX

SCR

FTI

VeloC

Native

= checkpoint library